How long a learned ranker's exposure gap lasts in a simulated matching market

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Abstract

When a learned ranker decides which candidates each side of a matching market sees, the concentration it adds to exposure is measured against an oracle that ranks by true utility, and the answer depends on when it is measured. In a simulated market of 1000 agents whose top-20 shortlists come from a neural collaborative filter, the ranker's exposure Gini exceeds the oracle's by +0.313 at round 6 and by +0.0253 at round 99, and each of 40 seeds then enters and stays inside a pre-registered margin of 0.002, between rounds 194 and 269. At round 0 the untrained ranker gives every agent on a side nearly the same twenty candidates. Across training doses on fresh logs the length of the transient is set by optimizer steps rather than market rounds (elasticity −0.99), but a warm start and a larger market do not follow that rule, and a ranker trained on data from a market it does not control closes the gap slightly faster. Because the ranker is trained on the oracle's own lists, the gap is the rate at which an imitator converges. Over 1000 rounds an exponential with a half-life of 29.6 rounds fits best, and the last fifty rounds average +0.00046, above zero and inside the margin. When a share of agents is replaced every round the gap had not closed by round 600: at 0.5, 1 and 2 percent per round it is +0.016, +0.040 and +0.101 over rounds 500 to 599, still falling at the lowest rate and rising at the other two. A companion institutional comparison at the shipped gate changes sign between a five-round horizon and a late one, and at an inert gate it does not.

Code and data. All code, data and run records are at `https://github.com/freddyscheung-067/hypergamic-matching-markets` and are archived on Zenodo under the concept DOI 10.5281/zenodo.23620786, which resolves to every released version; this paper's results are those of release v2.0.0.

Scope. The model is a simulation with synthetic agents. It makes no claim about real people or any real platform, and it uses no human data.

1 Introduction

When a recommender sits between the two sides of a matching market, it decides which candidates anyone sees. An agent on nobody's shortlist is not rejected; it is never considered, and no preference data record the difference. The usual way to measure what the ranker adds is to compare exposure under the learned ranker with exposure under an oracle that ranks by true utility, on the same population and seed. The oracle's own concentration comes from preferences, since agents that many others value highly appear on many lists; the difference between the two is concentration the ranker adds.

In the market simulated here that difference changes with the horizon at which it is read: large at first, small after a hundred rounds, and inside a margin of 0.002 Gini after three hundred. A comparison at one horizon cannot tell a lasting bias from a cost of the ranker's early training, and the two call for different readings of what the ranker does. This paper measures the whole trajectory, how it ends, and what sets its length.

The convergence objection. The feedback-loop literature leads one to expect the opposite. A system that learns from data its own choices generate can lock in [Ensign et al., 2018, Chaney et al., 2018, Mansoury et al., 2020], and a ranker that shows only the agents it already rates highly might never learn about the others. The answer in this model is a property of its design, stated here rather than discovered later: the ranker's training data are not generated by its own shortlists. Each round's log holds every agent's true top admissible candidates and uniform negatives, so the ranker sees informative data about everyone, whether or not it showed them. Once the gate stops binding, those positives are the oracle's own top-k lists, so the ranker is trained to imitate the benchmark it is scored against, and the gap measures how fast that imitation converges. Its shortlists reach that log through one channel only: they decide who receives proposals, which moves acceptance thresholds, which changes admissible sets. §5.4 measures how much that channel matters, and §7 says what changes in a market whose recommendations are its only data.

Contributions.

1. **A transient with a measured length.** On 40 seeds the gap peaks at $+0.313$ near round 6, and every seed enters and stays inside a pre-registered equivalence margin of 0.002 between rounds 194 and 269. Over 1000 rounds an exponential decay predicts held-out rounds better than a power law or an exponential to a constant, with a half-life of 29.6 rounds and a residual of $+0.00046$, inside the margin and above zero (§5.1, §5.2).

2. **What ends it.** At round 0 the untrained ranker's scores are almost all a per-candidate term, so every agent on a side sees nearly the same twenty candidates and the exposure Gini is the closed form $1 - 2k/N$. The gap closes as that common list breaks into each agent's own top twenty. Across training doses on fresh logs its length is set by optimizer steps, not rounds (elasticity -0.99); a ranker trained on another market's data closes it slightly faster; and pretraining removes about four fifths of the decay on the registered measure and less on the other three (§5.4).

3. **An open market.** When a share of agents is replaced every round, the gap had not closed by round 600: over rounds 500 to 599 it is $+0.016$, $+0.040$ and $+0.101$ at 0.5, 1 and 2 percent per round, still falling at the lowest rate and rising at the other two. At the two higher rates newcomers are under-exposed and long-tenured agents over-exposed relative to the oracle (§5.5).

4. **An institutional comparison that depends on the horizon.** Read at five rounds, no setting of a small complete-information subgraph beats the oracle-ranked market; read after the transient, a range of subgraph sizes at the lowest proposal cost beats it, and beats the learned-ranker market too (§6).

5. **Sensitivity of single trajectories.** A 10^{-7} perturbation to the initial weights grows to the size of the between-seed spread, so a reported number estimates a property of the seed distribution, not of one run (§4).

What the model is about. It is a two-sided matching market with a learned ranker. One side's utility includes a bonus for a partner whose status exceeds its own by a margin δ. The exposure results do not depend on it: across $\delta \in \{0, 0.5, 1\}$ the peak gap moves by 0.010 and the median entry round by 12 rounds, against 0.494 and 68 rounds across the chemistry axis (§5.3). The status bonus is the one asymmetry between the sides, so the welfare comparison of §6 may depend on it; that dependence was not tested. Every parameter is a modeling choice and nothing is calibrated to data (§7).

2 Related work

Feedback loops and long-run fairness. A system that learns from data its own decisions produce can mislead itself. Ensign et al. [2018] prove that predictive policing sends police back to the same areas regardless of the true rate when discovered incidents feed the next allocation. Sinha et al. [2016] recover the ratings a recommender has not influenced by deconvolving its feedback loop, and Schnabel et al. [2016] treat recommendations as treatments to obtain unbiased estimators under selection. Chaney et al. [2018] show by simulation that training on algorithmically confounded data homogenizes behavior without raising utility, Mansoury et al. [2020] that a feedback loop amplifies popularity bias and hits minority-group users hardest, and Jiang et al. [2019] separate the echo chamber from the filter bubble and propose ways to slow degeneration. D'Amour et al. [2020] argue that static or single-step analyses miss long-term consequences and make simulation the instrument for them; Liu et al. [2018] show that even one step of feedback can turn a fairness criterion harmful. Perdomo et al. [2020] give the frame this paper fits: retraining on a distribution the model's own deployment shapes, and conditions under which it converges to a performatively stable point. Here the ranker retrains every round on data its past shortlists shape weakly, and the gap is the path to that point. The claim that one horizon can mislead is D'Amour et al. [2020]'s; what is added is a measurement of the transient for a learned ranker in a matching market, of its length, and of what sets it.

Exposure fairness. Singh and Joachims [2018] define fair rankings by constraints on the exposure each item receives, and Biega et al. [2018] amortize attention across a sequence of rankings so individual position bias evens out. Both design a ranking to meet an exposure target. Here exposure is the number of shortlists an agent appears on, and the gap is measured against an oracle, not a target: the oracle's own Gini of about 0.647 is concentration that comes from preferences, which no fair-exposure constraint was meant to remove. Fleder and Hosanagar [2009] show that recommenders built on sales and ratings can reduce sales diversity, a one-sided analog of the concentration measured here.

Reciprocal recommenders and matching platforms. Pizzato et al. [2010] introduce the reciprocal recommender for online dating, Tomita et al. [2022] describe a matching-theory recommender deployed on a dating platform, and Chen et al. [2023] run a matching-model ranker in a field experiment and find less recommendation inequality and more matches. Kanoria and Saban [2021] show that a platform restricting what agents can do can raise welfare, Halaburda et al. [2018] that a platform can compete by limiting choice, Arnosti et al. [2021] how design can cut wasted search and screening, and Rios et al. [2021] how a national admissions clearinghouse was designed and run. Mladenov et al. [2020] show that matching each user to the best provider performs poorly when providers need engagement to stay viable. Patro et al. [2020] give a two-sided fairness criterion for recommendations on a platform, and Stoica et al. [2018] show how a recommender changes the

visibility of groups in a growing network. Each of these designs or evaluates a ranker or a platform; none of those cited here measures how long a learned ranker's effect on exposure takes to settle.

Matching theory. Deferred acceptance finds a stable matching [Gale and Shapley, 1962]; the proposer-optimal stable matching is the acceptor-pessimal one, and no procedure is both stable and strategy-proof [Roth, 1982, Knuth, 1996]. Roth and Rothblum [1999] show that truncating a list changes outcomes, Ashlagi et al. [2017] that a slight imbalance makes the stable matching essentially unique, and Akbarpour et al. [2020] study markets in which agents arrive and depart. Under a compatibility constraint that is nested in a scalar status, the minimal unmatched mass follows from Hall's marriage theorem [Hall, 1935] in its deficiency form as a supremum over thresholds. Table 3 checks the simulator against these results. A learned top-k visibility constraint is not nested, and no closed form is known for it.

Hypergamy as an economic object. Saint-Paul [2015] derives hypergamous matching from a trade-off in which women forgo more attractive mating opportunities to credibly signal legitimate children; Almås et al. [2023] use parental earnings rank as a predetermined measure and find hypergamy an important feature of matching in Norway; Esteve et al. [2016] show the reversal of the educational gender gap moving marriage toward hypogamy with no change in preferences required. This paper's status bonus is a stylized version of the preference these works study, and nothing here tests them.

Simulation methodology. Windrum et al. [2007] and Fagiolo et al. [2007] set out why validating an agent-based model is a family of practices rather than a single test. This paper claims verification against closed forms, not validation against data. Its equivalence test is the two one-sided tests procedure of Lakens [2017].

3 Model

The market has N agents, $N/2$ on each side: a proposing side M and an accepting side W. It runs in discrete *rounds*. In each round a learned ranker gives every agent a shortlist of k agents from the other side, and a truncated deferred-acceptance pass clears the market over those shortlists and nothing else. Throughout, a *round* is one pass of the market loop of Algorithm 1, and a *training pass* is one sweep of the ranker's optimizer over one round's training log. The shipped configuration has $N = 1000$, $k = 20$ and one training pass per round. Table 1 lists every parameter with its shipped and swept values.

3.1 Agents and utility

Agent i carries four traits $x_i = (s_i, a_i, c_i, t_i)$: status, attractiveness, sociability and age, drawn independently as

$$s_i = e^{0.8Z}, \quad a_i = [0.5 + 0.15Z]_0^1, \quad c_i = [0.5 + 0.15Z]_0^1, \quad t_i = 30 + 6Z, \tag{1}$$

with a fresh standard normal Z for each draw and $[\cdot]_0^1$ clipping to the unit interval. Each trait is standardized against the whole population, so $z(s_i) = (s_i - \bar{s})/\hat{\sigma}_s$ with the sample mean and the sample standard deviation over all N agents. Status is standardized on its own lognormal scale, so $z(s)$ is right-skewed. Agent i also carries three taste multipliers $m_i^\alpha, m_i^\beta, m_i^\gamma$, independent lognormal with log-mean $-\sigma_\tau^2/2$ and log-sd σ_τ, so each has mean one.

The utility of i for a candidate j on the other side is

$$U_{ij} = m_i^{\alpha}\alpha\, z(s_j) + m_i^{\beta}\beta\, z(a_j) - m_i^{\gamma}\gamma\, d_{ij} + \lambda\, \mathbf{1}[i \in W]\, \mathbf{1}[z(s_j) \geq z(s_i) + \delta] + \sigma_L L_{ij}, \tag{2}$$

where $d_{ij} = \|z(x_i) - z(x_j)\|_2$ is the distance between the standardized trait vectors. The fourth term is the status-surplus bonus: an acceptor gains λ from a candidate whose standardized status exceeds her own by at least δ. It is the only asymmetry between the sides, and it is one variant of the utility rather than the subject of the paper (§5.3). The last term is persistent pairwise chemistry: L_{ij} is a standard logistic draw computed from a hash of the run seed and two per-agent keys, fixed for the life of the pair and asymmetric ($L_{ij} \neq L_{ji}$). Its standard deviation is $\sigma_L\pi/\sqrt{3}$, which at the shipped $\sigma_L = 1.5$ is 2.72 utility units.

Agents also carry an intent $\tau_i \sim \text{Beta}(2,2)$, a values score $g_i \sim \text{Beta}(2,2)$ and an eligibility flag for the institutional comparison. They enter only that comparison (§6, Appendix A) and change no exposure quantity.

3.2 Acceptance thresholds and the gate

Each agent has an acceptance threshold θ_i in utility units, and its *admissible set* is $A_i = \{j : U_{ij} \geq \theta_i\}$. At round 0, θ_i is the n_A-th largest of i's utilities over the other side, with $n_A = \text{round}(\phi N/2)$, so every agent starts with exactly n_A admissible candidates. After each round,

$$\theta_i \leftarrow \theta_i + \eta\left(\frac{m_i}{p_i} - \theta_i\right) \quad \text{if } p_i > 0, \tag{3}$$

where p_i is the number of proposals i made (on M) or received (on W) in that round's matching, $m_i \in \{0,1\}$ is whether i was matched, and $\eta = 0.1$. An agent with no proposal keeps its threshold. The yield m_i/p_i lies in $[0,1]$ while the initial threshold is the tenth-highest utility an agent has, far above one, so the update lowers θ_i geometrically toward the unit interval for every agent who takes part in the matching.

The threshold can change what an agent sees only if the admissible set is shorter than the shortlist. The candidate set is the other side, of size $N/2$, so $|A_i| = \phi N/2$ at round 0 and the gate binds then exactly when

$$\phi < \frac{2k}{N}. \tag{4}$$

At $N = 1000$ and $k = 20$ this is $\phi < 0.04$; the shipped $\phi = 0.02$ gives ten admissible candidates against a shortlist of twenty. As thresholds fall the admissible sets grow past k and the gate stops binding, within a few rounds for agents who propose or are proposed to.

3.3 The training log

Each round the ranker trains on a log built afresh from the current population and thresholds. For every agent i on either side:

- the *positives* are the $\min(k, |A_i|)$ members of A_i with the highest U_{ij}, each labeled 1;
- the *negatives* are kn_- candidates drawn uniformly without replacement from the rest of the other side, each labeled 0, with $n_- = 4$.

A positive is determined by the utility and the threshold; there is no randomness in it. The negative draw for agent i in round r comes from a generator seeded with (run seed, i, k, r), so every round has fresh negatives and the log does not depend on the thread count. Logs are not accumulated:

the ranker sees one round's log at a time, and what it retains is in its weights. At the shipped values an agent contributes at most $k(1 + n_-) = 100$ rows to a log, and 99 on average over a run.

The log does not depend on the ranker's shortlists. Its positives are exactly the shortlists an oracle with the same thresholds would hand out, and its negatives are uniform. The ranker's own choices reach its training data through one channel only: they decide who proposes to whom, which moves thresholds by (3), which changes admissible sets. This is the fact that answers the convergence objection of §1, and §5.4 tests it. It also means that, once the gate stops binding, the positives are the oracle's top-k lists by true utility: the ranker imitates the benchmark that the gap is measured against, and the gap is the rate at which that imitation converges. It is not the exposure of a ranker trained on engagement that its own lists produce.

3.4 The ranker

The ranker is neural collaborative filtering in the fused form of He et al. [2017]: a generalized matrix factorization branch and a multilayer perceptron branch over one id space, in which every agent is both a user and a candidate. Each branch has its own user and item embeddings of dimension 64. The perceptron maps the concatenated pair embedding through layers of width 128, 64 and 32 with ReLU activations, and a linear output layer reads the element-wise product of the factorization embeddings beside the perceptron's output to give a logit $\hat{y}_{uv}$. Embeddings are initialized $N(0, 0.01^2)$ and dense layers with the library default. Training minimizes binary cross-entropy on the logits with positive weight 4, using Adam [Kingma and Ba, 2015] at learning rate 10^{-3} and weight decay 10^{-5}, in batches of 4096 rows shuffled by a seeded generator. The weights and the optimizer state persist across rounds. One training pass over a round's log is about 25 optimizer steps at the shipped values; the dose experiment of §5.4 changes the number of passes.

Agent u's shortlist S_u is the k candidates on the other side with the highest $\hat{y}_{uv}$. It is *not* restricted to A_u: the ranker ranks the whole other side. An agent's *exposure* e_j in a round is the number of shortlists that contain it, $0 \le e_j \le N/2$.

The architecture is deliberately standard. Rendle et al. [2020] show that a tuned dot product beats this learned similarity at recommendation accuracy; no result here depends on the ranker being a good one, only on its being learned from the log above, and §7 returns to what a different ranker could change.

3.5 Clearing

The matching is deferred acceptance [Gale and Shapley, 1962] with M proposing, run on the shortlists: a proposer works down his own shortlist, and an acceptor holds the best proposal from among the proposers *on her shortlist* and rejects everyone else, including proposers she has never been shown. The result is stable with respect to the shortlists and in general not with respect to true preferences (Table 3). With M proposing it is the proposer-optimal stable matching of the truncated lists and the acceptor-pessimal one [Roth, 1982, Knuth, 1996]. Every arm uses the same proposing side, so the comparisons below are between rankers, not between directions.

3.6 Comparison arms

Four rules produce shortlists on the same population, utilities, seed and matching code; only the rule differs, and each arm runs its own threshold dynamics.

- **NCF**: the learned ranker above.

- **Oracle**: the top k of A_i by true utility, shorter when $|A_i| < k$. It is the benchmark a learned ranker falls short of, not an achievable institution: it knows the chemistry term, which nothing observable reveals.
- **Random**: $\min(k, |A_i|)$ members of A_i drawn uniformly without replacement, from a generator seeded with (run seed, round).
- **Popularity**: one list per side, the k agents of the other side with the highest exposure in the previous round, ties to the lower index. In round 0, which has no history, it ranks by true utility.

3.7 Metrics

All metrics are per round. With exposures $e_1, \ldots, e_N$ over all agents of both sides, sorted ascending as $e_{(1)} \le \cdots \le e_{(N)}$:

$$G = \frac{2\sum_{i=1}^{N} i\, e_{(i)}}{N \sum_{i=1}^{N} e_{(i)}} - \frac{N+1}{N} \qquad \text{exposure Gini,} \tag{5}$$

$$R = \frac{1}{N} \left|\{j : e_j > 0\}\right| \qquad \text{reach,}$$

$$\Delta = G_{\text{NCF}} - G_{\text{oracle}} \qquad \text{the gap, same seed and round.}$$

The gap is the paper's quantity. It measures concentration in excess of the oracle's, and the oracle's own Gini is near 0.647 (§5.1): that much concentration comes from preferences, because agents with high status and attractiveness are on many agents' true top-k. A positive gap is concentration the ranker adds to that, not concentration as such.

Two more metrics are reported. The *clearing rate* is $2\mu/N$ for μ matched pairs. *Algorithmic loss* is $(W^* - W)/W^*$, where $W = \sum(U_{ij} + U_{ji})$ over matched pairs and W^* is the same sum for deferred acceptance on complete true-preference lists. In the fully concentrated market, where every list on a side is the same k agents, exactly k pairs can form and the clearing rate is $2k/N$; when lists spread it is larger (Supplementary note S1).

3.8 An open market

Every run except those of §5.5 has a closed population: no agent arrives or leaves. In the open-market runs a share ν of the N slots changes hands at the start of every round after the first. The departing agents are drawn uniformly at random, so tenure is geometric with mean $1/\nu$ rounds, and each is replaced in the same slot by a fresh agent: new traits, tastes, chemistry key, intent and values from the distributions of §3.1; a threshold calibrated against the market it enters; the ranker's four embedding rows for the slot re-drawn from the initial distribution, with the shared layers kept. This is replacement, not net arrival: the market's size is fixed. The departures and the newcomers depend only on the run seed and the round, so every arm on a seed sees the same population at every round.

The code also supports generational turnover, in which the least-matched fifth of the population is replaced at fixed intervals. No experiment reported here uses it.

3.9 The second environment

The institutional comparison of §6 sets the market above, Environment A, against Environment B: a subgraph of the same population in which every member sees every other member. An agent

Algorithm 1 One round r of the market. $\mathcal{L}$ is the round's training log.

if $r > 0$ and $\nu > 0$ **then** replace a share ν of the slots (§3.8)
end if
$\mathcal{L} \leftarrow$ positives and negatives for every agent (§3.3)
take ρ training passes of the ranker over $\mathcal{L}$ ($\rho = 1$ shipped)
for each agent u **do**
    $S_u \leftarrow$ the top k of the other side by score
end for
match by deferred acceptance, M proposing, on $\{S_u\}$ (§3.5)
record exposure e, G, R, the clearing rate and welfare
for each agent i with $p_i > 0$ **do**
    $\theta_i \leftarrow \theta_i + \eta\,(m_i/p_i - \theta_i)$
end for

is eligible for B if its eligibility flag is set and $g_i \geq g_{\min} = 0.3$; a share ϕ_B of each side is drawn into B, and the sides are balanced by dropping the excess. Inside B the shortlist is the oracle rule restricted to members, and proposing costs c_B. An agent i proposes to j only if

$$\tau_i \, U_{ij} \, p > c_B, \tag{6}$$

where p is the acceptance rate measured on the previous round's log for i's side (0.5 before any measurement). A matched pair survives the round with probability

$$\sigma_{ij} = \sigma_0 (\tau_i \tau_j)^{\sigma_\alpha} g_{ij}^{\sigma_\beta}, \qquad g_{ij} = 1 - |g_i - g_j|, \tag{7}$$

with $\sigma_0 = 1$ and $\sigma_\alpha = \sigma_\beta = 0.5$. When a pair dissolves the member with lower intent defects; the partner loses $L(1 - \tau_{\text{def}})$ with $L = 1$, and in B the defector is charged $d = 0.5$. In A the same survival draw applies with $d = 0$ and no proposal cost. Welfare in a round is $\sum(\tau_i U_{ij} + \tau_j U_{ji})$ over matched pairs, less proposal costs, less dissolution losses. Agents B excludes are scored at the payoff they would get in A with B's matched agents removed from A's market, so that excluding an agent cannot raise B's welfare by itself.

4 Design

Every experiment runs the model of §3 at the shipped configuration except where it says otherwise, at one thread, from a committed tree, and writes each seed's data beside a record of the commit, the full configuration and its hash, and the library versions. Table 2 lists the experiments.

Seeds. There are two sets. The *exploratory* set, seeds 42 to 61, contains seeds 42 to 46, on which every headline of the project's first release rested. The *confirmation* set, seeds 100 to 119, took no part in any earlier choice. The confirmation set is primary and the exploratory set replicates it; experiments with ten seeds use 100 to 109 and those with five use 100 to 104. A seed fixes a population (traits, tastes, chemistry, intents and values) and every random stream downstream of it. The arms of an experiment share each seed's population, so every comparison is a paired per-seed difference.

Symbol	Meaning	Shipped	Swept
N	agents	1000	500, 2000
k	shortlist length	20	not swept
ϕ	admissible share at round 0	0.02	$20/N$ with N; 0.2 (inert gate)
η	threshold step	0.1	not swept
ν	replacement share per round	0	0.005, 0.01, 0.02
α	status weight	1	not swept
β	attractiveness weight	2	1, 3
γ	distance penalty	0.3	not swept
λ	status-surplus bonus	1.5	not swept
δ	status margin (sd)	0.5	0, 1
σ_L	chemistry scale	1.5	0.5, 3.0
σ_τ	taste spread (log-sd)	0.5	0.2, 1.0
s	status, lognormal (log-mean, log-sd)	0, 0.8	not swept
a	attractiveness, normal, clipped	0.5, 0.15	not swept
c	sociability, normal, clipped	0.5, 0.15	not swept
t	age, normal	30, 6	not swept
τ, g	intent, values: Beta	(2, 2)	not swept
	embedding dimension	64	not swept
	perceptron widths	128, 64, 32	not swept
	learning rate, weight decay	10^{-3}, 10^{-5}	not swept
	positive weight, negatives per positive	4, 4	not swept
ρ	training passes per round	1	0.5, 2, 4
	exploration share	0	0.05, 0.2
	warm-start passes	0	50
ϕ_B	subgraph share	0.5	0.02 to 1.00 (ten levels)
c_B	proposal cost	0.25	0.01, 0.05, 0.10, 0.25
$g_{\min}$	values filter	0.3	not swept
d	defection penalty	0.5	not swept
$\sigma_0, \sigma_\alpha, \sigma_\beta$	survival	1, 0.5, 0.5	not swept
L	partner loss	1	not swept

Table 1: Every parameter of the model, with its value in the shipped configuration and the values swept. All are modeling choices: none is estimated from data. The swept values are those of the sensitivity sweep (§5.3), the dose, exploration and population-size experiments (§5.4) and the open market (§5.5).

Rounds, passes and the margin. A *round* is one pass of Algorithm 1; a *training pass* is one sweep of the optimizer over a round's log, about 25 optimizer steps at the shipped values. Most experiments run 300 rounds; the floor test runs 1000 and the open market 600. The *margin* is 0.002 Gini. A seed's gap is *inside the margin* in a round when its absolute value is below 0.002, and the seed's *entry round* is the first round from which its gap stays inside the margin to the end of the run. A seed outside the margin at the last round is *censored*. An entry round differs from the first round a seed is inside the margin whenever the gap re-crosses the margin after first touching it; only the entry round is used.

Pre-registration. Each experiment's design, the quantity read, a prediction with a number in it and a kill condition were committed before the experiment ran, at commit `480cd6c` of the repository's pre-registration file. Four departures are labeled where their results appear: the primary warm-start measure was fixed after the candidate numbers had been computed on earlier data (Supplementary note S4), and a comparator check for exploration (§5.4), a trend test for the open

Experiment	Question	Seeds	Rounds	Commit
R2-E9	the transient: four arms, 300 rounds	40	300	`349b484`
R2-E10	floor or slow decay: 1000 rounds	10	1000	`f4d2b8c`
R2-E11	one-knob sweep, eight cells	10	300	`349b484`
R2-E13a	training dose: 0.5, 2, 4 passes per round	10	300	`f4d2b8c`
R2-E13b	exploration slots	10	300	`f4d2b8c`
R2-E13b-check	comparator check for exploration (post hoc)	10	300	`9b11a36`
R2-E13c	off-policy ranker	10	300	`f4d2b8c`
R2-E13d	warm start	40	300	`349b484`
R2-E14	open market: replacement	10	600	`f4d2b8c`
R2-E15	institutional grid, rounds 250-299	10	300	`f4d2b8c`
R2-E15i	institutional slice, inert gate	10	5	`f4d2b8c`
R2-E15s	institutional grid, rounds 0-4 (post hoc)	10	5	`ec0eb83`
R2-E16	floating-point sensitivity, ten runs per seed	5	300	`9b11a36`
R2-E17	population size 500 and 2000	10	300	`f4d2b8c`
R2-H1r	H1 at the shipped gate, round-0 log	10	1	`f4d2b8c`
R2-H5r	H5 at the shipped gate	10	40	`f4d2b8c`

Table 2: The experiments, one row per registration, with the commit every seed of it ran from. Seeds 100–119 are the confirmation set and 42–61 the exploratory set (§4). Every experiment was registered before it ran except the two rows marked post hoc; a third post-hoc check, a trend test, needed no new run. R2-E12, the diagnostics of §5.1, reads quantities recorded in R2-E9's runs and has no row of its own. Source: `data/processed/horizon/run_index.csv`, built from the per-seed run records.

market (§5.5) and the five-round institutional grid (§6) were added after the registered results were read. Table 7 lists every registered prediction and whether it held, scored from the result files by a script rather than by reading.

Statistics. Intervals are 95% Student-t intervals over seeds unless stated. The replicate is the seed, never the round: the rounds of one run are one trajectory. BCa bootstrap intervals (10,000 resamples) accompany the headline round-99 gap as a check, and at twenty seeds their bounds differ from the t bounds by at most 0.0007. Equivalence is tested by two one-sided tests against ± 0.002 [Lakens, 2017], whose interval is the 90% t interval by construction. Benjamini–Hochberg at $q = 0.05$ corrects the forty cells of the institutional grid, separately for each comparison arm.

Why one run is not a measurement. A perturbation of 10^{-7} to every initial weight of the ranker, applied ten times to each of five seeds, leaves the first rounds unchanged and first moves the gap at round 2. The spread over a seed's ten runs then grows to 0.62 of the spread across twenty seeds at round 99 and to 1.25 at round 299 (Figure 1). A single run's late-round gap is as much a property of its floating-point history as of its population. Every number here is therefore a mean over seeds with an interval over seeds, and every run is at one thread, because the thread count changes the order of floating-point reductions and so is part of a run's identity.

Verification. Table 3 checks the matching and exposure machinery against results known exactly: the deficit identity that follows from Hall's theorem under a nested constraint and its Gaussian closed form, uniqueness of the stable matching under common rankings, the popularity ranker's closed form, and stability of truncated deferred acceptance on its own lists. Agreement shows the machinery is correct. It says nothing about whether the model describes a population.

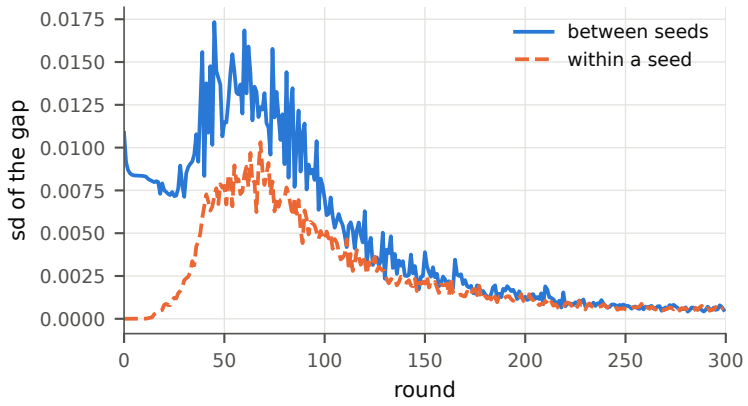


Figure 1: Standard deviation of the gap between seeds (twenty confirmation seeds) and within a seed (ten runs whose initial weights differ by 10^{-7}, averaged over five seeds), by round. Source: `data/processed/horizon/e16_fp/` and `e09_canonical/`.

Check	N	Measured	Evidence	Verdict
Deficit identity (Hall)	$N = 300$	exact on 5 seeds	matching = $\sup_t$ formula	agrees
Deficit, uniform vs. Beta(2,1)	$N = 20{,}000$	0.2519 vs 1/4	single seed, tol 0.02	agrees
Gaussian closed form, equal variances	$N = 5{,}000$	0.3864	formula 0.3829	agrees
Exact supremum, unequal variances	$N = 5{,}000$	0.4135	supremum 0.4031	agrees
Common rankings: unique stable matching	$N = 300$	same matching both ways	0 blocking pairs, 5 seeds	agrees
Perturbed rankings	$N = 300$	22 / 8 / 11 / 12 / 8 pairs differ	uniqueness fails, stability holds	agrees
Popularity ranker: Gini $1 - 2k/N$	$N = 500$, 1000, 2000	error below 10^{-12}	Gini and reach, $k = 20$	agrees
Truncated DA on top-k lists	$N = 300$	0 blocking pairs on the lists	blocking pairs remain against full preferences	stable on its lists

Table 3: Verification against exact results. Rows 1–4: the minimal unmatched share under a nested compatibility constraint, by exact maximum matching, against the supremum formula that follows from Hall's theorem [Hall, 1935], and against its Gaussian closed form. Rows 5–6: deferred acceptance under common rankings and under rankings perturbed by 1% noise [Gale and Shapley, 1962, Knuth, 1996]. Row 7: the popularity ranker's exposure Gini against $1 - 2k/N$. Row 8: blocking pairs of truncated deferred acceptance against its own lists and against full preferences. Source: `data/processed/rigor/validation.csv`, written by `scripts/run_validation.py`; a test fails if the file differs from a fresh computation.

5 Results

5.1 The transient

Figure 2(a) shows the gap over 300 rounds. On the confirmation set it starts at +0.269, rises to +0.313 at round 6 and falls to +0.0253 [+0.0217, +0.0290] at round 99, positive on all twenty seeds; the exploratory set peaks at +0.310 and is +0.0225 [+0.0187, +0.0263] at round 99. Every one of the forty seeds then enters the margin and stays inside it, and none is censored. The entry round has a median of 227 (range 199–261) on the confirmation set and 222.5 (194–269) on the exploratory set.

Over rounds 250 to 299 the mean gap is +0.00004 [−0.00014, +0.00023], positive on 11 of twenty seeds. The two one-sided tests reject at $t = 23.2$ and 22.3 on 19 degrees of freedom, and the 90% interval [−0.00011, +0.00019] lies inside the margin; the exploratory set gives +0.00005 [−0.00013, +0.00023] and the same verdict. The registered predictions for this experiment all held (Table 7), including that seeds 42 to 46 reproduce an earlier 300-round run of the same configuration to every printed digit.

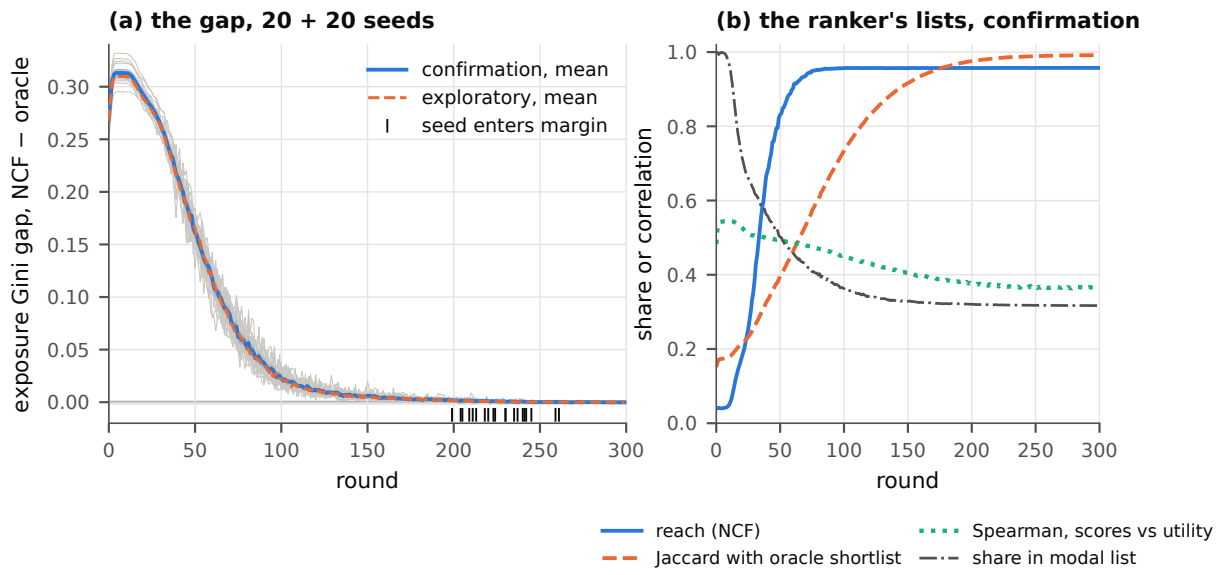


Figure 2: (a) The gap, NCF minus oracle exposure Gini on the same seed and round: the confirmation-set mean (20 seeds, 95% t band), the exploratory-set mean (20 seeds), each confirmation seed (thin gray), and each confirmation seed's entry round (ticks). The shaded strip is the ± 0.002 margin. (b) Diagnostics of the ranker on the confirmation set: reach; mean Jaccard overlap of an agent's shortlist with the oracle's; mean Spearman correlation between an agent's scores and its true utilities over the whole other side; mean share of an agent's shortlist inside its side's modal list. Source: `data/processed/horizon/e09_canonical/`.

The other arms bound the gap (Figure 3). The oracle's exposure Gini settles at 0.647 within 5 rounds, as thresholds relax, and does not move after that: this is the concentration preferences produce. The NCF arm's Gini falls from the popularity closed form, $1-2k/N = 0.96$, to the oracle's level. At round 0 the oracle, random and popularity arms coincide, because the gate admits fewer than k candidates and every arm hands out all of them. From round 1 the popularity arm sits at 0.96, and the random arm falls to 0.352. The peak gap is therefore nearly a closed form: the ranker's $1 - 2k/N = 0.96$ minus the oracle's settled 0.647, about 0.313. What the experiments measure is how long the ranker takes to leave the common list, not the size of the peak.

Exposure converges; welfare does not. At round 299 the ranker's algorithmic loss is 0.618, against 0.593 for the oracle, a paired difference of $+0.026$ $[+0.018, +0.033]$ over twenty seeds. The ranker's shortlists agree with the oracle's as sets (mean Jaccard overlap 0.992) while its scores over the whole other side have a rank correlation of only 0.37 with true utility. Exposure counts whether an agent is on a list and not where, and deferred acceptance reads the order, so the remaining welfare difference is consistent with lists that hold the right agents in a different order. Within-list order was not measured directly.

5.2 A floor, or a slow decay?

A tail inside the margin at round 300 is also what a slow decay would look like at that horizon, so ten confirmation seeds were run to round 999. Their first 300 rounds reproduce the R2-E9 runs exactly. Three forms were fitted to the seed-mean gap from its peak to round 500 and judged by how well they predict rounds 501 to 999, which they did not see (Figure 4). The pure exponential

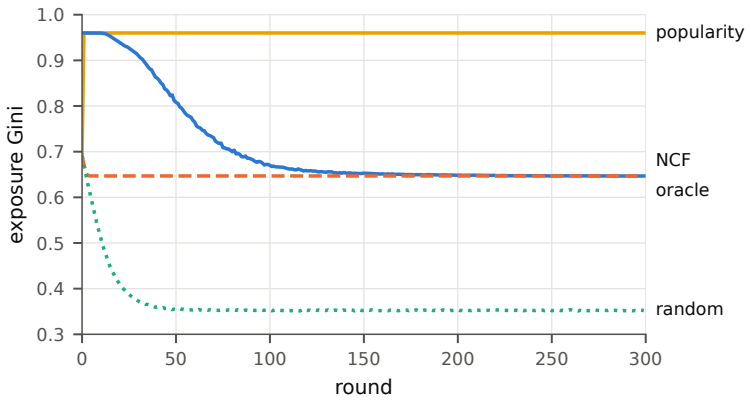


Figure 3: Exposure Gini of the four arms over 300 rounds, mean over the twenty confirmation seeds. Source: `data/processed/horizon/e09_canonical/`.

predicts best, with a root-mean-square error of 0.00056 against 0.0034 for an exponential to a constant and 0.016 for a power law. Its half-life is 29.6 rounds, 95% interval 28.8–30.9 from 2000 resamples of the seeds. The power law is not a candidate: it predicts a gap of 0.014 at round 999, several times the margin. The exponential to a constant fits the first 500 rounds better by AIC and BIC, but its constant is -0.0029 $[-0.0032, -0.0028]$, a floor below zero that the held-out rounds contradict.

Over rounds 950 to 999 the gap is $+0.00046$ $[+0.00013, +0.00080]$, positive on 9 of ten seeds, and the 90% interval $[+0.00019, +0.00073]$ lies inside the margin. Its change between rounds 900 to 949 and 950 to 999 is -0.000015 $[-0.000247, +0.000216]$, so the series has stopped moving. By the registered definition this is not a positive floor, which needed the tail interval to lie above $+0.002$. It is a residual of 23 percent of the margin, inside the range the margin declares of no consequence, and above zero at ten seeds. The oracle's Gini is constant to every printed digit from round 300 on, so the residual is a small rise in the ranker's concentration late in training. Nothing below depends on whether it is real.

5.3 Where it holds

One knob at a time. Eight cells each move one modeling knob from its shipped value: taste spread σ_τ to 0.2 and 1.0, chemistry noise σ_L to 0.5 and 3.0, the weight ratio β/α to 1 and 3, and the status margin δ to 0 and 1, with ten seeds each over 300 rounds. The shipped cell is seeds 100 to 109 of the canonical run (Figure 5, Table 4). The five claims were re-registered as trajectory claims before the cells ran.

The peak gap is positive with an interval excluding zero in all nine cells (C1t), and its size ranges from $+0.060$ at chemistry noise 0.5 to $+0.554$ at 3.0. Near the peak the ranker is still close to the popularity closed form, so the peak gap is mostly the distance from 0.96 down to the oracle's Gini, and that distance grows as chemistry noise makes preferences less common: the oracle's Gini at round 299 is 0.900 at noise 0.5 and 0.406 at 3.0. The tail passes the equivalence test in seven of nine cells (C2t). It fails at chemistry noise 3.0, where the peak is largest and the tail is $+0.0019$, with 2 of ten seeds still outside the margin at round 299 and a 90% interval reaching $+0.0031$; and at chemistry noise 0.5 for the opposite reason: the tail is -0.0019, negative with an interval excluding zero (C5t), and the ranker ends less concentrated than the oracle. The weight ratio $\beta/\alpha = 3$ also ends with a negative tail, -0.0009, inside the margin; the registration had predicted

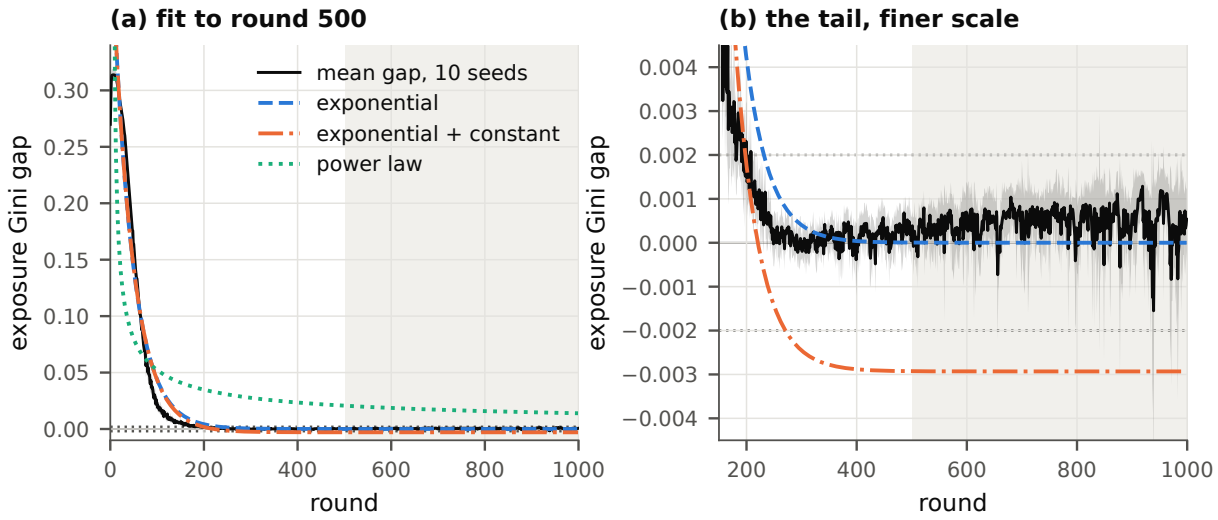


Figure 4: The gap over 1000 rounds, ten confirmation seeds (mean, 95% t band), with three forms fitted from the peak to round 500 and evaluated on rounds 501 to 999 (shaded). Dotted lines: the ± 0.002 margin. (b) is the tail of (a) at a finer scale. Source: `data/processed/horizon/e10_long/`.

a reversal at chemistry noise 0.5 only. Reach at round 299 is at least 0.75 in eight cells (C3t) and 0.502 at chemistry noise 0.5, where the oracle's reach is 0.502: with little idiosyncratic taste many agents are on no one's true top twenty. Clearing is non-degenerate in every cell (C4t). The transient occurs in every cell; its size and length are set by how common preferences are.

Population size. At $N = 500$ and $N = 2000$ the shortlist stays at $k = 20$ and the gate target is set to $\phi = 20/N$, so every agent again starts with ten admissible candidates and the gate binds; ten seeds each (Table 5). The round-0 Gini is the closed form at each size. The peak gap is $+0.324$, $+0.314$ and $+0.300$ at $N = 500$, 1000 and 2000, falling slightly where the registration predicted a rise. The first round from which the gap stays below 0.05, $T_{0.05}$, has a mean of 95, 87 and 136 rounds, and at $N = 2000$ 7 of ten seeds are still outside the margin at round 299. The transient is longest in the largest market, and it is not monotone in N.

The horizon in data units. A length in rounds is a unit of the model. Every agent's log holds about 99 rows per round at every N, at most twenty of them positives, so a length in observations per agent is proportional to a length in rounds and gives the same ratios across N. In optimizer steps $T_{0.05}$ is 1,188, 2,135 and 6,580 at the three sizes, because a pass over a larger market takes more steps. Neither unit makes the length independent of market size, so a transient length measured here does not transfer to another market in rounds, in data or in steps. Table 5 gives $T_{0.05}$ in all three units for every experiment that has one.

5.4 What ends it

The common list breaks up. At round 0 a per-candidate term explains at least 0.9999 of the within-agent variation in the untrained ranker's scores on every one of the forty seeds, at least 0.990 of every agent's shortlist is its side's modal list, and the exposure Gini is within 2.5×10^{-5} of the popularity closed form (Figure 2(b)). Every agent on a side sees nearly the same twenty

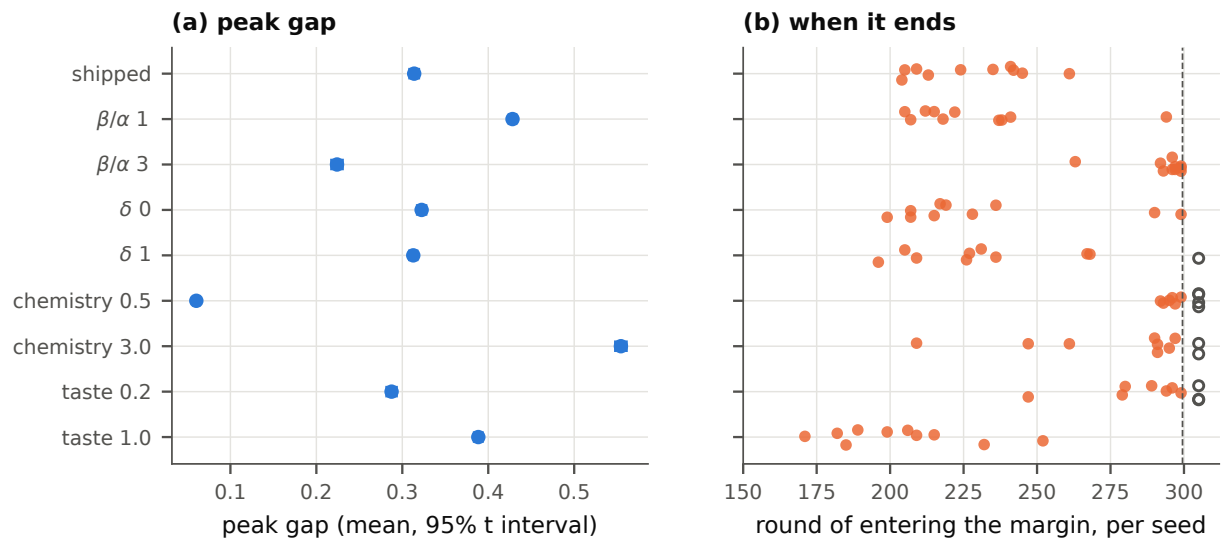


Figure 5: The one-knob sweep, ten seeds per cell. (a) Peak gap, mean with 95% t interval. (b) Each seed's entry round; open markers at the right edge are seeds still outside the margin at round 299. Source: `data/processed/horizon/e11_sweep/` and, for the shipped cell, seeds 100–109 of `e09_canonical/`.

candidates. As training proceeds the share of identical lists falls below 0.1 by round 15, the per-candidate term's share of score variation falls below one half by round 135, and the mean overlap of an agent's shortlist with the oracle's rises from 0.151 to 0.992.

The registered prediction tied the gap to ranking quality measured as the rank correlation between an agent's scores and its utilities over the whole other side. It failed in sign: across rounds 6 to 299 the gap and that correlation are positively correlated on every seed (median +0.87, minimum +0.80), because the correlation falls as the gap closes. The ranker starts by ordering every agent's candidates by one shared score, which agrees moderately with each agent's own utility because the candidates many agents value share high status and attractiveness. It then learns which twenty candidates each agent ranks highest, and its training labels separate those twenty from uniform negatives without ordering the rest. The gap is a quantity about the top twenty, and among these diagnostics it is the overlap with the oracle's shortlist and the breakup of the modal list that move with it.

Optimizer steps, not rounds. Training at 0.5, 1, 2 and 4 passes per round, ten seeds each, the gap stays below 0.05 from round 174, 87, 44 and 22 on average: doubling the passes halves the length (Figure 6(a)). In cumulative optimizer steps the four curves fall on one (Figure 6(b)), and the step count at that threshold is between 2,133 and 2,190 at every dose. The elasticity of the length in rounds with respect to the dose is -0.99 $[-1.01, -0.98]$, as registered. Each round's log draws on nearly the same distribution as the last, so what the transient takes is optimizer steps. At four passes the gap overshoots: it ends at -0.0028, below the margin, with 5 of ten seeds outside it at round 299. More training eventually leaves this ranker less concentrated than the oracle, and this experiment does not say why.

Cell	Peak gap	Entry	Censored	Tail $\times 10^3$	C1t	C2t	C3t	C4t	C5t
shipped	0.314	230	0	+0.12	yes	yes	yes	yes	no
$\sigma_\tau = 0.2$	0.288	295	3	-0.15	yes	yes	yes	yes	no
$\sigma_\tau = 1.0$	0.388	202	0	-0.12	yes	yes	yes	yes	no
$\sigma_L = 0.5$	0.060	298	4	-1.85	yes	**no**	**no**	yes	yes
$\sigma_L = 3.0$	0.554	291	2	+1.95	yes	**no**	yes	yes	no
$\beta/\alpha = 1$	0.428	220	0	+0.07	yes	yes	yes	yes	no
$\beta/\alpha = 3$	0.224	296	0	-0.91	yes	yes	yes	yes	yes
$\delta = 0$	0.322	218	0	-0.15	yes	yes	yes	yes	no
$\delta = 1$	0.313	229	1	+0.17	yes	yes	yes	yes	no

Table 4: The sweep's five trajectory claims, registered before the cells ran. C1t: peak gap positive, t interval excluding zero. C2t: two one-sided tests on the rounds 250–299 tail pass at ± 0.002. C3t: reach at round 299 at least 0.75. C4t: clearing rate at round 299 with an interval excluding zero. C5t: tail mean negative with an interval excluding zero (a reversal). Ten seeds per cell. Source: `data/processed/horizon/e11_sweep/analysis/cells.csv`.

Feedback through thresholds. A second ranker, trained each round on the log of a market it does not control (one run by the oracle rule or by the random rule), starts from the same weights and, at round 0, the same data as the market's own ranker; the thresholds are what differ after that. At round 49 its gap is smaller than the market ranker's by 0.024 $[0.015, 0.034]$ on the oracle market's data and by 0.020 $[0.008, 0.032]$ on the random market's (Figure 7(a)). The registered prediction, a difference under 0.01, failed. The effect is about 13 percent of the gap at that round, about 4 rounds of decay at that point of the curve, and it is gone by round 299. The mechanism is the one channel of §3.3: concentrated shortlists leave many acceptors without proposals, their thresholds stay high, and their logged positives stay truncated for longer. Feedback delays the end of the transient; it does not sustain it.

Exploration. Replacing the last m entries of every shortlist with candidates drawn uniformly ($m = 1$ or 4 of 20) lowers the whole-list gap by construction. The quantity of interest is the gap of the exploiting part, the first $20 - m$ entries against the oracle's first $20 - m$. Its difference from the non-exploring ranker's gap is -0.014 at round 49 and -0.016 at round 99 at $m = 1$, and -0.009 and -0.032 at $m = 4$; every interval excludes zero except the one at round 49 for $m = 4$, and the registered prediction of a difference inside ± 0.01 fails (Figure 7(b)). The likely route is the same: random slots spread proposals and so relax more thresholds. At round 299 the exploiting part ends above the control, +0.007 at $m = 1$. Cutting a list by utility rank and by score rank differs even when the two lists hold the same agents, so a check added after these results were read compares the exploring ranker's head with a non-exploring ranker's head of the same length. On that comparison the round-299 excess is +0.0024 $[-0.0017, +0.0065]$, an interval that includes zero, so the late excess was the comparator's; the faster closing is not: the differences at rounds 49 and 99 are -0.016 and -0.010 at $m = 1$ and -0.016 and -0.014 at $m = 4$, every interval excluding zero.

An informed start. Pretraining the ranker for fifty passes on the round-0 log before the market opens raises round-0 reach from 0.041 to 0.866 and cuts the round-0 gap from +0.268 to +0.060 (forty seeds). On the registered measure, the warm arm's fall from round 0 to round 299 as a share of the control's, 20.8 percent [17.9, 23.6] of the decay survives on the confirmation set and 24.0 percent on the exploratory set: the informed start removes about four fifths of it (Table 6).

Experiment	Condition	Rounds	Rows per agent	Optimizer steps	Censored
R2-E9	shipped, confirmation	87	8,512	2,131	0 of 20
R2-E9	shipped, exploratory	86	8,395	2,102	0 of 20
R2-E11	$\beta/\alpha = 1$	93	9,142	2,287	0 of 10
R2-E11	$\beta/\alpha = 3$	81	7,849	1,968	0 of 10
R2-E11	$\delta = 0$	85	8,338	2,088	0 of 10
R2-E11	$\delta = 1$	88	8,615	2,158	0 of 10
R2-E11	$\sigma_L = 0.5$	26	2,458	618	0 of 10
R2-E11	$\sigma_L = 3.0$	113	11,204	2,803	0 of 10
R2-E11	$\sigma_\tau = 0.2$	85	8,278	2,075	0 of 10
R2-E11	$\sigma_\tau = 1.0$	87	8,539	2,136	0 of 10
R2-E13a	0.5 passes per round	174	17,067	2,190	0 of 10
R2-E13a	2 passes per round	44	4,260	2,133	0 of 10
R2-E13a	4 passes per round	22	2,142	2,144	0 of 10
R2-E13c	off-policy, oracle market's data	75	not logged	1,865	0 of 10
R2-E13c	off-policy, random market's data	75	not logged	1,878	0 of 10
R2-E13d	warm start, confirmation	62	6,146	2,642	0 of 20
R2-E13d	warm start, exploratory	63	6,210	2,659	0 of 20
R2-E17	N = 500	95	9,296	1,188	0 of 10
R2-E17	N = 2000	136	13,377	6,580	0 of 10

Table 5: $T_{0.05}$, the first round from which the gap stays below 0.05, in three units: rounds, rows of training log per agent, and optimizer steps. Mean over seeds; censored seeds are excluded and counted. Sources: `data/processed/horizon/` (`e09_canonical`, `e11_sweep`, `e13a_dose`, `e13c_offpolicy`, `e13d_warm`, `e17_size`).

Quantity	Basis	Confirmation (100–119)	Exploratory (42–61)
gap (primary)	round 0	20.8% [17.9, 23.6]	24.0% [21.9, 26.1]
level	round 0	32.0% [29.4, 34.6]	34.4% [32.7, 36.1]
gap	own peak	69.1% [67.8, 70.4]	68.3% [67.1, 69.4]
level	own peak	69.3% [68.0, 70.6]	68.6% [67.4, 69.8]

Table 6: Share of the control arm's decay that survives a warm start, four constructions, each computed per seed and averaged, 95% t interval over twenty seeds per set. Round-0 basis: (warm_0 − warm_{299}) / (control_0 − control_{299}); peak basis: each arm from its own maximum. Gap: NCF minus oracle Gini; level: NCF Gini. The registered primary is the gap on the round-0 basis. Source: `data/processed/horizon/e13d_warm/analysis/summary.csv`.

The registered upper bound of one half held; the registered round-0 reach above 0.8 held on every seed but one (0.774). The share is larger on levels, because the oracle's Gini also falls in both arms, and near 0.7 measured from each arm's own peak, because the warm arm climbs for a few rounds from its informed start before it decays. By round 299 the two arms differ by +0.0003 on the confirmation set. Fifty passes over one log add optimizer steps without adding data, and they are not worth as much as steps on fresh logs: the warm arm's gap stays below 0.05 from round 62 against 87 for the control, sooner in rounds, but after 2,642 optimizer steps against 2,131 (Table 5). The dose result's steps are steps on a new log each round.

5.5 An open market

In every run so far the population is fixed, so the ranker eventually learns every agent. When a share ν of the slots is replaced every round by fresh agents with new traits, new embeddings and calibrated thresholds (§3.8), some agents are always new. Over rounds 500 to 599 the gap is $+0.016$ $[+0.015, +0.018]$ at $\nu = 0.5$ percent per round, $+0.040$ $[+0.037, +0.044]$ at 1 percent and

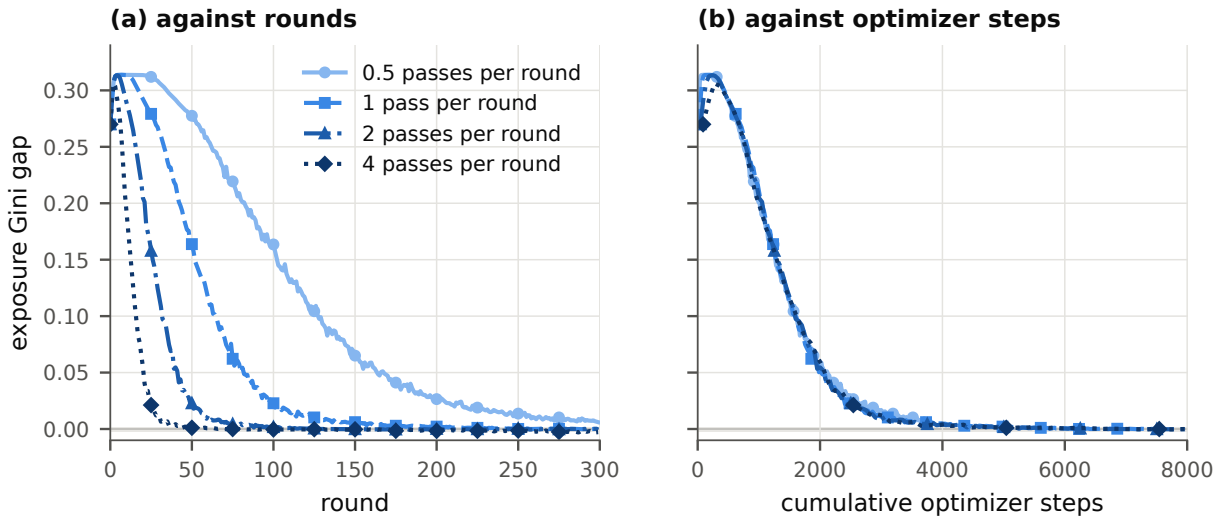


Figure 6: The gap at four training doses, ten seeds each (mean), against rounds (a) and against cumulative optimizer steps (b). Shaded strip: the ±0.002 margin. Source: `data/processed/horizon/e13a_dose/` and, for one pass, seeds 100–109 of `e09_canonical/`.

+0.101 [+0.096, +0.106] at 2 percent, positive on every seed and increasing in the rate, as registered (Figure 8(a), ten seeds per rate). At every rate it is above the margin.

The registered stationarity check fails at every rate. Its paired comparison of rounds 590–599 with 580–589 finds no change ($p = 0.13$, 0.71 and 0.08), but its second criterion, a largest round-to-round step in the seed-mean series below a tenth of the seed standard deviation, fails: the step is 1.03, 0.56 and 0.76 of it. That criterion was written for the closed market, whose series stops moving; here every round's arrivals move every round's mean. A trend test added after this was read finds the gap still moving over rounds 400 to 599: it changes by −0.0029 [−0.0051, −0.0008] at 0.5 percent per round, +0.0040 [+0.0007, +0.0073] at 1 percent and +0.0196 [+0.0110, +0.0283] at 2 percent, every interval excluding zero. At the lowest rate the gap is still falling, so its level over rounds 500 to 599 overstates the gap at the end of the run; at the two higher rates it is rising. None of the three levels is a steady state at 600 rounds. What survives is the weaker statement the trend allows: at every rate the gap stays above the margin to the end of the run, and it is larger at a higher rate.

By tenure (Figure 8(b)), agents in their first ten rounds receive a share of the ranker's exposure below their share of the population at 1 and 2 percent per round (0.917 and 0.827 of proportional), while the oracle gives them close to their share (0.992 and 1.001), and agents with fifty rounds or more receive more than their share (1.008, 1.057, 1.120). At 0.5 percent per round the newest agents are over-exposed instead (1.091), which fails the registered prediction that they are under-exposed at every rate.

The ordering follows from a ratio. An agent stays for $1/\nu$ rounds on average, 200, 100 and 50 rounds at the three rates, against a closed-market entry round of about 227, so a higher rate leaves a larger share of agents newer than the ranker's learning time. No rate with $1/\nu$ well above that time was run, so the experiment shows that the gap tracks this ratio and does not locate where it falls inside the margin.

A round has no calendar length here, so a replacement rate per round cannot be read as a platform's churn without knowing how many rounds of retraining a platform's period contains.

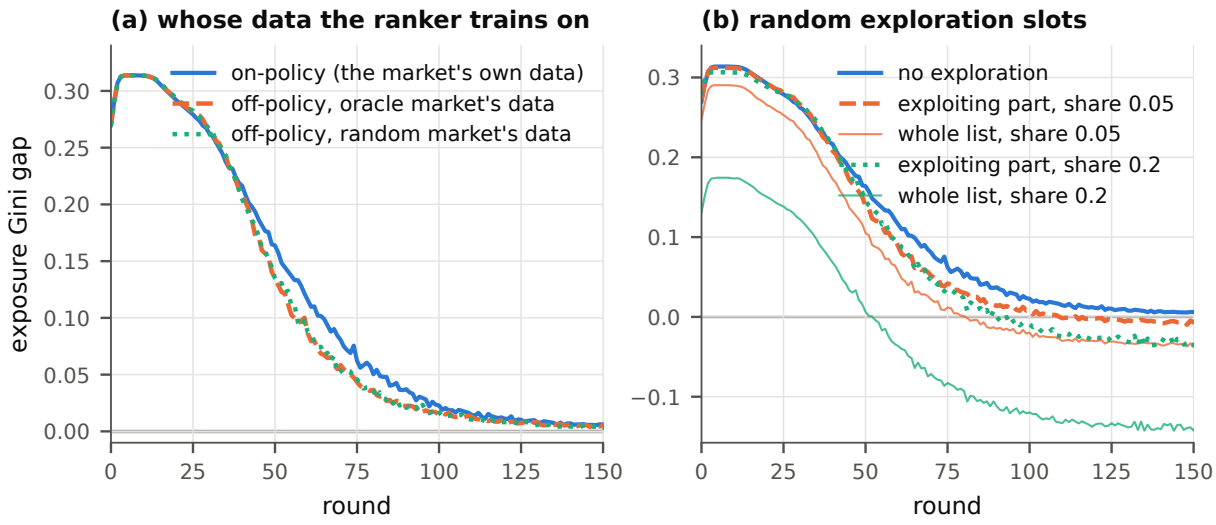


Figure 7: (a) The gap of a ranker trained on its own market's data (on-policy) and of a ranker trained on the data of a market run by the oracle or the random rule (off-policy), ten seeds, mean. (b) With exploration slots: the gap of the exploiting part of each list and of the whole list, against the non-exploring ranker, ten seeds, mean. Source: `data/processed/horizon/e13c_offpolicy/`, `e13b_explore/`, `e09_canonical/`.

What the open market shows is that the gap tracks the rate at which the ranker meets agents it has not learned, and that it stays above the margin at every rate tested.

6 The institutional comparison

Environment B (§3.9) is a subgraph of the same population in which every member sees every other and proposing costs c_B. It is compared with the mediated market, Environment A, over a grid of ten subgraph sizes and four costs, ten seeds per cell, against two versions of A: the oracle, and the learned ranker. Welfare is averaged over rounds 250 to 299, after the transient, and B's welfare is on the common basis that scores every agent B excludes at what A would give it (Figure 9).

At the lowest cost, $c_B = 0.01$, B beats A over a range of sizes. With Benjamini–Hochberg correction over the forty cells, 4 cells beat the oracle A, at sizes 0.35 to 0.80, and 6 beat the learned-ranker A, at sizes 0.25 to 1.00, all at $c_B = 0.01$. The largest difference is at size 0.65 against the oracle, $+58.3$ $[+18.6, +98.0]$, and at 0.65 against the learned ranker, $+99.3$ $[+55.4, +143.3]$. At $c_B \geq 0.05$ no cell beats A after correction, and B falls further behind as the cost and the subgraph grow. The margin over the learned ranker is the larger one because that A's own welfare is lower, 917.5 against 977.0, although its exposure has by then converged to the oracle's (§5.1). Why its welfare is lower was not tested; the reading offered in §5.1, the right agents in a different order, is consistent with the numbers and unmeasured.

The same grid read over its first five rounds, on the same seeds (a check added after the late-window result was read), finds no cell in which B beats the oracle A: every cell's mean is negative, and A beats B after correction in 28 of the forty. At size 0.65 and $c_B = 0.01$ the difference is -9.9 $[-25.9, +6.1]$ over rounds 0 to 4 and $+58.3$ over rounds 250 to 299. A comparison read at five rounds and one read after the transient therefore point in different directions, for a quantity that involves no learning on A's side once the oracle's thresholds have settled. This experiment shows

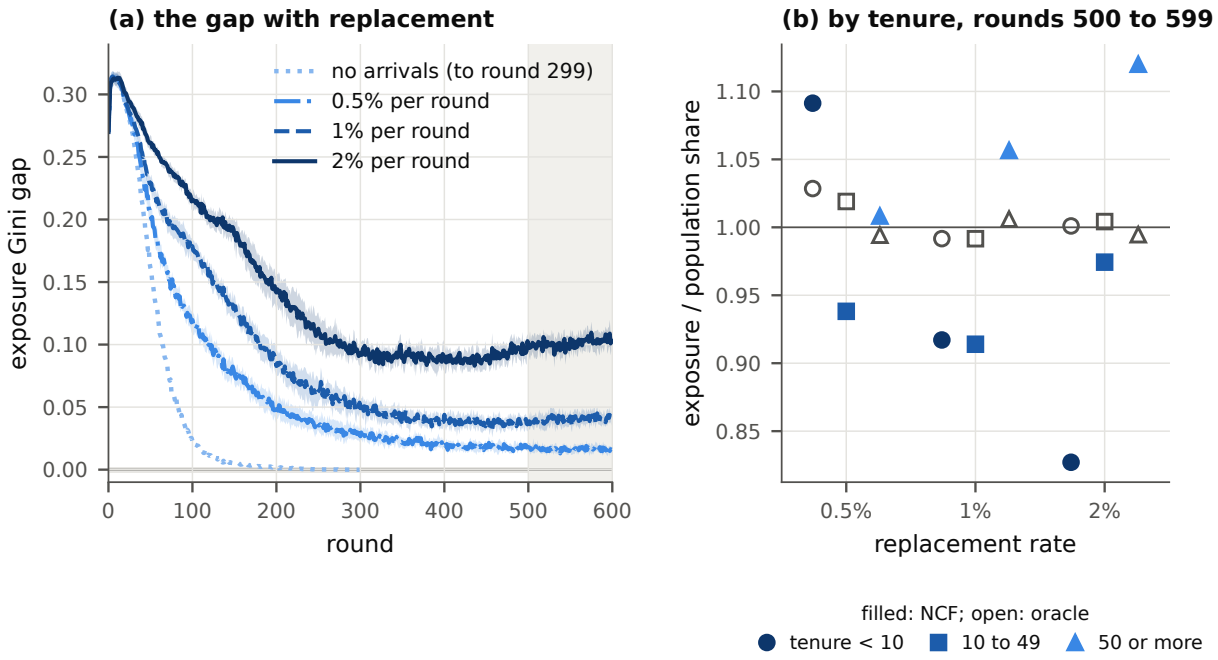


Figure 8: (a) The gap with a share of agents replaced each round, ten seeds per rate (mean, 95% t band), and the closed market (seeds 100–119, to round 299). Shaded: rounds 500 to 599, over which the levels are read. (b) Each tenure band's share of exposure divided by its share of agents, rounds 500 to 599: NCF (filled) and oracle (open). Source: `data/processed/horizon/e14_arrivals/`.

the difference and does not isolate its cause. In a gate regime that never binds ($\phi = 0.2$, read at five rounds, ten fresh seeds) size 0.65 ranks first of ten at $c_B = 0.01$, as registered, with a mean of $+56.6$ $[+16.5, +96.7]$, positive over the same five rounds in which the shipped gate gives -9.9. The sign of the early comparison thus depends on the gate as well as on the horizon, and the two were not varied together in one grid.

7 Limitations

Nothing is calibrated. Every parameter in Table 1 is a modeling choice: the utility weights, the trait distributions, the chemistry scale, the taste spread, the shortlist length and the gate target. The results describe this mechanism under a stated utility, not any population, and the size of the gap and the length of its transient move with chemistry noise and the weight ratio (§5.3).

The ranker's data are not its own. Each round's log holds every agent's true top admissible candidates, so the ranker learns about agents it never shows. On a platform whose only data are the responses to its own recommendations, the lock-in that the feedback-loop literature describes could occur, and nothing here rules it out. This is the largest difference between the model and a deployed system.

Exposure counts membership, not position. The gap measures whether an agent is on a list, while deferred acceptance reads the order of the list. That is why exposure converges while welfare

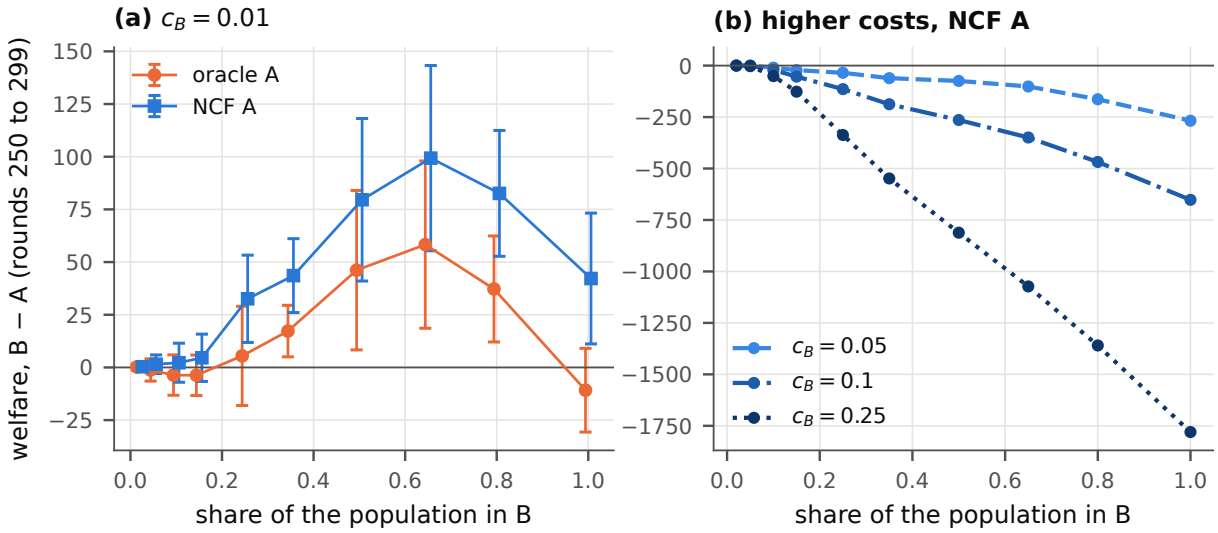


Figure 9: Welfare of Environment B minus Environment A over rounds 250 to 299, ten seeds per cell, on the common basis. (a) $c_B = 0.01$ against the oracle A and the learned-ranker A, mean with 95% t interval. (b) Higher costs, against the learned-ranker A, mean. Source: `data/processed/horizon/e15_institutional/`.

does not (§5.1). An exposure measure weighted by position could follow a different trajectory, and it was not measured.

The open market has not reached a steady state. At 600 rounds its gap is still falling at the lowest replacement rate and rising at the other two (§5.5), so its levels are readings at a horizon, not limits. Replacement is uniform and immediate; no agent leaves in response to an outcome.

One population size carries most results. The oracle needs every agent's full preference ranking, and the experiments run at $N = 1000$ with $N = 500$ and 2000 as checks, where the transient's length was not monotone in N.

There is no exit on outcomes. No agent leaves because it was unmatched or poorly matched, so there is no attrition margin, one hypothesis cannot be posed (Appendix A), and a mechanism that drove participants away would look free in the welfare comparison.

The benchmark is not achievable. The oracle ranks by true utility, including the pairwise chemistry no observable reveals, so every gap is a distance from a bound and not from an alternative a platform could deploy. Truncation is imposed rather than chosen: agents do not report or game their lists.

8 Conclusion

In a closed market the concentration a learned ranker adds to exposure is a cost of its training. It is set by the number of optimizer steps the ranker has taken, not by how long the market has run, and a comparison with an oracle at a fixed number of rounds reads the ranker's training state

as much as the ranker. An exposure audit of a learned ranker is therefore incomplete without the horizon it was read at, stated in steps.

In a market whose population turns over, the cost had not ended by round 600 at any rate tested. The gap is larger when the ranker meets agents it has not learned more often, stays above the margin, and at the higher rates falls on newcomers and favors long-tenured agents. Which of the two describes a platform depends on that rate, which this model cannot supply. Neither reading covers welfare: shortlists that hold the oracle's agents in another order give lower welfare, and the institutional comparison, read after the transient, finds a complete-information subgraph better than either version of the mediated market over a range of sizes.

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A The project's other hypotheses

The project began with six hypotheses, each with a kill condition. The exposure-gap hypothesis is the body of the paper and the institutional one is §6. The other four are below, each re-measured here at the shipped configuration where it can be measured at all.

H1: a cost on proposing selects on intent. If proposing to j costs c, agent i proposes only when $\tau_i U_{ij} p > c$, so the mean intent of proposers should rise with c. On the round-0 log at the shipped configuration, ten seeds, it rises from 0.498 at $c = 0$ to 0.657 at $c = 2$, and it is non-decreasing over the eleven costs, as registered. The kill condition, a flat curve, is not met (source: `data/processed/horizon/h1r/`).

H2: thresholds rise in defense against pooling. Not instrumented. The threshold update reads only an agent's proposal and match counts; it never reads the intent of proposals received, so no path exists by which the predicted relation could appear.

H5: an interaction between status drift and mediation. A factorial over exogenous status drift (on or off) and mediation (the learned ranker or the oracle), 40 rounds, welfare over the last ten, ten seeds. The interaction is -21.3 $[-47.9, +5.4]$, an interval that includes zero, as registered; the algorithmic term is -717.8 $[-747.3, -688.4]$ and the exogenous term -15.7 $[-35.8, +4.5]$. A 40-round horizon lies inside the transient, so the algorithmic term measures a ranker that is still training (source: `data/processed/horizon/h5r/`).

H6: high-intent agents leave a mediated market faster. Not instrumented. No agent leaves in response to an outcome, so differential attrition cannot be posed.

B Reproducibility

Code, data and build. The repository holds the C++ core, the Python layer, every configuration, every result file and this paper's sources. One command regenerates the numbers, tables and figures from the result files and compiles the paper: `python paper/build.py`; adding `-analysis` first reruns every analysis from the per-seed files. Release v2.0.0 is archived on Zenodo, under the concept DOI 10.5281/zenodo.23620786.

Pre-registration. The repository's pre-registration file is append-only, and each experiment is named by its entry there (R2-E9 and so on). Every experiment of Table 2 except the post-hoc rows was registered at commit `480cd6c`; the warm-start primary measure at `7dfcd8c`, written after the four candidate measures had been computed on earlier five-seed data, and saying so; the

three post-hoc checks at `b8e34e4` and `d91a312`; and one execution failure, before any of its data existed, at `9b11a36`. That failure was a configuration value written without a decimal point, which the configuration reader parsed as text, so all fifty floating-point-sensitivity runs stopped before computing anything and were rerun unchanged in design. Table 7 scores every registered prediction.

Thread count. Every run is at one thread. The ranker's training is not bitwise reproducible across thread counts, because the order of floating-point reductions changes, and §4 shows how far a difference of that size travels. A test fails if a configuration sets a thread count other than one for a quoted run.

Run records. Each seed of each experiment wrote a record of the commit it ran from, whether that tree was clean, the full configuration and its hash with the seed removed, the library versions and the wall time. The analysis scripts refuse a directory whose records disagree on commit or configuration hash. Table 2 is built from these records.

Archived directories. Results produced before the experiments of Table 2 are kept in the repository and labeled archival; no number in this paper comes from them. The 300-round run of the shipped configuration in the project's first release does not regenerate from the current code at any thread count from 1 to 24, in its NCF arm only, and that release's twenty-seed 100-round run differs from a fresh run in the fourth decimal. A later 300-round five-seed run at one thread reproduces exactly: seeds 42 to 46 of the canonical run match it to every printed digit. The earlier institutional grids, warm-start runs and sweep were read at horizons or seed counts that the experiments of Table 2 replace.

C Supplementary notes

S1. The clearing ceiling of a common list. If every proposer's list is the same k acceptors, only those k acceptors receive proposals, and each holds at most one, so at most k pairs form and the clearing rate is at most $2k/N$. The same holds when every acceptor's list is the same k proposers. Lists that differ across agents remove the bound. At round 0, when the ranker hands every agent on a side nearly the same list, the learned ranker's clearing rate is 0.040; at round 299 it is 0.259, against $2k/N = 0.04$.

S2. Five seeds against twenty. Every headline of the project's first release rested on seeds 42 to 46. On the exploratory set of the canonical run, seeds 42 to 61 at round 99, the standard deviation over those five seeds is 0.0083 for the gap against 0.0082 over all twenty, and 0.0045 for algorithmic loss against 0.0222. Ordered from the smallest standard deviation, the block 42 to 46 has rank 12 of the sixteen contiguous five-seed blocks for the gap and rank 1 for algorithmic loss: an interval on algorithmic loss built on those five seeds alone would have been about 5.0 times too narrow. Every interval in this paper is computed over ten or twenty seeds.

S3. Two interval constructions. At twenty seeds the BCa bootstrap interval for the round-99 gap is $[+0.0224, +0.0292]$ against the t interval $[+0.0217, +0.0290]$. The two agree to 0.0007, which is why the paper reports t intervals throughout and BCa as a check.

Experiment	Registered prediction	Measured	Held
R2-E9	peak gap in [0.27, 0.34] at a round in 3-10	0.313 at 6; 0.310 at 6	yes
R2-E9	tail equivalent to zero (TOST at 0.002), both sets	passes both	yes
R2-E9	at least 15 of 20 seeds enter the margin by round 299	20 and 20 of 20	yes
R2-E9	round-99 gap in [0.015, 0.035]	0.0253; 0.0225	yes
R2-E9	seeds 42-46 reproduce the earlier one-thread run exactly	max difference 0	yes
R2-E12	round-0 Gini within 0.002 of 0.96, all 40 seeds	max deviation 2.5e-05	yes
R2-E12	round-0 reach within 0.002 of 0.04, all 40 seeds	max deviation 0.003	**no**
R2-E12	round-0 modal share above 0.95 and item share above 0.9	min 0.990 and 0.9999	yes
R2-E12	gap-Spearman correlation below -0.8 on 36 of 40 seeds	0 of 40 (median +0.87)	**no**
R2-E12	identical share below 0.1 by round 100	rounds 15, 15	yes
R2-E12	item share below 0.5 by round 100	rounds 135, 131	**no**
R2-E12	rank correlation of round and Jaccard above 0.9	1.000; 1.000	yes
R2-E11	C1t holds in all nine cells	9 of 9	yes
R2-E11	C2t holds in at least seven of nine	7 of 9	yes
R2-E11	C3t fails at chemistry 0.5 and nowhere else	fails at chemistry 0.5	yes
R2-E11	C4t holds in all nine cells	9 of 9	yes
R2-E11	C5t (a reversal) at chemistry 0.5 and nowhere else	at beta/alpha 3, chemistry 0.5	**no**
R2-E13d	round-0 reach under the warm start above 0.8, every seed	minimum 0.774	**no**
R2-E13d	primary measure's upper bound below 0.5, both sets	0.236; 0.261	yes
R2-E13d	both peak-basis measures above 0.5, both sets	min 0.683	yes
R2-E10	no positive floor	none	yes
R2-E10	tail (rounds 950-999) equivalent to zero	passes	yes
R2-E13a	elasticity of T(0.05) in the dose at most -0.75	-0.994	yes
R2-E13b	exploiting-part gap within the band of the control's	2 of 8 checks inside	**no**
R2-E13b	whole-list gap below the control's at round 20, share 0.2	upper bound -0.139	yes
R2-E13c	off-policy gap within the band of the on-policy gap	5 of 8 checks inside	**no**
R2-E15	no cell has B beating the oracle A	4 cells	**no**
R2-E15	at least as many cells beat NCF A as beat the oracle A	6 against 4	yes
R2-E15i	size 0.65 ranks in the top three of ten, inert gate	rank 1	yes
R2-H1r	mean intent of proposers non-decreasing in the cost	non-decreasing	yes
R2-H5r	interaction interval includes zero	[-47.9, +5.4]	yes
R2-E14	gap over rounds 500-599 increases with the rate	$0.0165 < 0.0403 < 0.1011$	yes
R2-E14	at rate 0.02 the interval excludes zero	lower bound +0.0959	yes
R2-E14	newest band under-exposed relative to the oracle at every rate	2 of 3 rates	**no**
R2-E16	within-seed spread at round 0 below 0.001	0	yes
R2-E16	within/between ratio reaches 0.3 in rounds 10-150	max 1.22	yes
R2-E17	round-0 Gini within 0.003 of 1 - 2k/N at each N	max 2.9e-04	yes
R2-E17	peak gap increases with N	0.324 , 0.314 , 0.300	**no**
R2-E17	T(0.05) within a factor of two between N = 500 and 2000	95 and 136 rounds	yes
Held			29 of 39

Table 7: Every registered prediction and whether it held, scored by `scripts/analyze_extras.py`.

S4. The warm-start measure. The share of decay surviving a warm start can be computed on levels or gaps, from round 0 or from each arm's peak. The version of this paper on the repository's main branch before this release printed exposure Gini levels under the name of gaps and quoted a level, peak-basis share. The primary measure here, the gap on the round-0 basis, was fixed after all four candidates had been computed on earlier five-seed data, and the registration says so. The four constructions differ for two reasons. On levels, the oracle's own Gini falls between round 0 and the end in both arms alike, and a level difference counts that fall in numerator and denominator. On a peak basis, the warm arm climbs for a few rounds after its informed start, and measuring it from that later and lower peak discards the part of the effect the pretraining produced. The four values on forty seeds are in Table 6.